

Course Title	Probability & Random Processes	Course Code	EE2011
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	Credit Hrs.	3
Knowledge Area	Electrical Engineering Fundamental (KA06)	Grading Scheme	Relative
HEC Knowledge Area	Engineering Foundation	Applicable From	Fall 2023
Pre-requisite(s)			

Course Objective	To understand the basic concepts and tools of probability and random variables and to apply them for the analysis of electrical engineering problems.
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No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Analyze a probabilistic experiment and investigate probabilistic systems	A1,Q1, M1	C4	1
2	Interpret the concept of random variables, probability distribution/density functions and expected values of a random variable	A2, Q2, M2	C3	1
3	Appraise prevalent discrete and continuous random variables and their distributions	A3, Q3, F	C4	1

Text Books	Title	Probability and Statistics for Engineers and Scientists 9th ed.
	Author	R. E. Walpole, R. H. Myers, S.L. Myers and Keying Ye
	Title	Probability and Random Processes for Electrical Engineers
	Author	Alberto Leon- Garcia, 2nd Edition
	Publisher	Pearson Education Inc, 1994
Reference Books	Title	Introduction to Probability and Statistics: 2nd Edition
	Author	Dimitri P. Bertsekas and John N. Tsitsiklis
	Publisher	

Week	Course Contents/Topics	Chapter*	CLO*
1	Introduction, sample spaces and events, axioms of Probability	1,2	1
2	Determining cardinality of sample space and events and assigning probabilities.	2	1
3	Additive rules, conditional probability	2	1
4	Independence and the multiplication rule and total probability	2	1
5	Bayes Theorem	2	1
6	Random variables, discrete probability function, cumulative distribution function (CDF)	3	2
7	Continuous random variable, probability distribution function (PDF)	3	2
8	Joint distribution function, Functions of Random Variables	3	2
9	Expectation, Variances	4	2
10	Expectation, Variances	4	2
11	Bernoulli Random Variable, Binomial Random Variable and Negative Binomial Distribution	5	3
12	Geometric Distributions, Hyper-geometric Distributions and Poisson Distribution	5	3
13-14	Uniform Random Variable, Normal Distributions, Exponential distribution	6	3
14-15	Gamma, Beta Distribution, Chi-Squared, Beta, Lognormal Distribution	6	3

*Reference book chapters are given in brackets

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterm (I+II)	30.0%
Final Exam	50.0%